

DS5500 Data Science Capstone

Skincare Product Recommendation System

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1 Introduction

1.1 Background

Skincare product recommendation systems have become increasingly popular, aiming to guide consumers through a vast and often overwhelming market. Traditional recommendation approaches frequently rely on collaborative filtering or content-based methods that primarily utilize user reviews, ratings, or product descriptions. While these methods can be effective, they often suffer from limitations such as subjectivity in user feedback, cold-start problems for new products, and a lack of transparency regarding *why* a particular product is recommended. There is a growing need for more objective, personalized, and explainable recommendation systems that can leverage the detailed chemical composition of products and individual skin characteristics.

1.2 Problem Statement

The core problem addressed by this project is the difficulty in providing accurate and personalized skincare product recommendations that go beyond superficial attributes. Consumers often struggle to identify products that are truly effective for their specific skin concerns due to the complex interplay of ingredients and individual skin biology. This project aims to solve this by developing a system that can intelligently match individual skin conditions with the efficacy of cosmetic ingredients. Specifically, the problem involves:

- **Machine Learning Challenge:** Developing models capable of processing and understanding complex, sequential ingredient lists and correlating them with product effectiveness and skin compatibility.
- **Data Integration:** Effectively combining disparate datasets (product information and ingredient knowledge bases) to create a comprehensive representation suitable for advanced recommendation algorithms.
- **Personalization and Explainability:** Creating a system that not only recommends products tailored to specific skin types and concerns but also provides clear explanations for these recommendations based on ingredient properties.

Solving this problem is significant as it can lead to more effective skincare routines, reduce consumer trial-and-error, and foster greater trust in product recommendations. It has relevance in both academic research, particularly in the application of deep learning and natural language processing (NLP) to consumer product data, and in industry, by offering a novel approach to product discovery and personalized marketing.

1.3 Objectives

The primary objectives of this project are to:

- **Ingredient Analysis:** Develop a methodology to process cosmetic ingredient lists as sequential data, utilizing deep learning techniques to extract meaningful product attributes and understand ingredient functions.
- **Personalized Recommendations:** Create a system that can effectively match detected skin concerns (input by the user) with the known efficacy and properties of ingredients to generate a ranked list of suitable products.
- **Explainability:** Ensure that the recommendation system provides transparent justifications for its suggestions, clearly indicating which ingredients in a recommended product address specific user-identified skin concerns.

1.4 Scope

This project focuses on the development and implementation of a deep learning-based skincare product recommendation system. The scope includes:

- Utilizing publicly available datasets from Kaggle, specifically a Sephora skincare product dataset and an INCI (International Nomenclature of Cosmetic Ingredients) ingredient list dataset.
- Implementing a **FAISS (Facebook AI Similarity Search)** vector database to enable efficient semantic search and retrieval of products based on their ingredient profiles and descriptions.
- Developing an interactive **Streamlit application** to demonstrate the recommendation system's functionality, allowing users to input skin concerns and receive personalized product suggestions.

Excluded from the current scope are real-time facial skin condition analysis (user input for skin concerns is assumed), and direct integration with external e-commerce platforms for purchasing. The project primarily focuses on the recommendation engine and its user interface for demonstration purposes.

2 Methodology

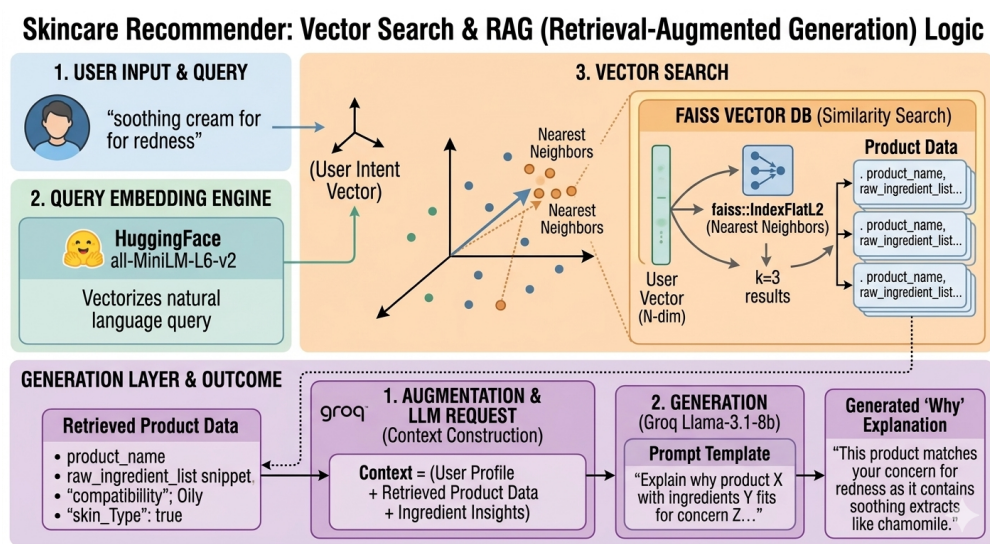


Figure 1: Methodology

2.1 Purpose of the Methodology

The chosen methodology aims to overcome the limitations of traditional skincare recommendation systems by focusing on a data-driven, ingredient-centric approach. By employing deep learning and vector database technologies, the project seeks to provide highly personalized, accurate, and explainable product recommendations. The rationale behind this approach is rooted in the belief that a deeper understanding of product ingredients and their interactions with various skin conditions is crucial for effective skincare guidance. This methodology allows for:

- **Objective Analysis:** Moving beyond subjective user reviews to analyze the intrinsic properties and functions of ingredients.
- **Enhanced Personalization:** Tailoring recommendations based on a detailed match between user-specified skin concerns and ingredient efficacy.
- **Transparency:** Providing clear justifications for recommendations, thereby building user trust and educating them about product formulations.

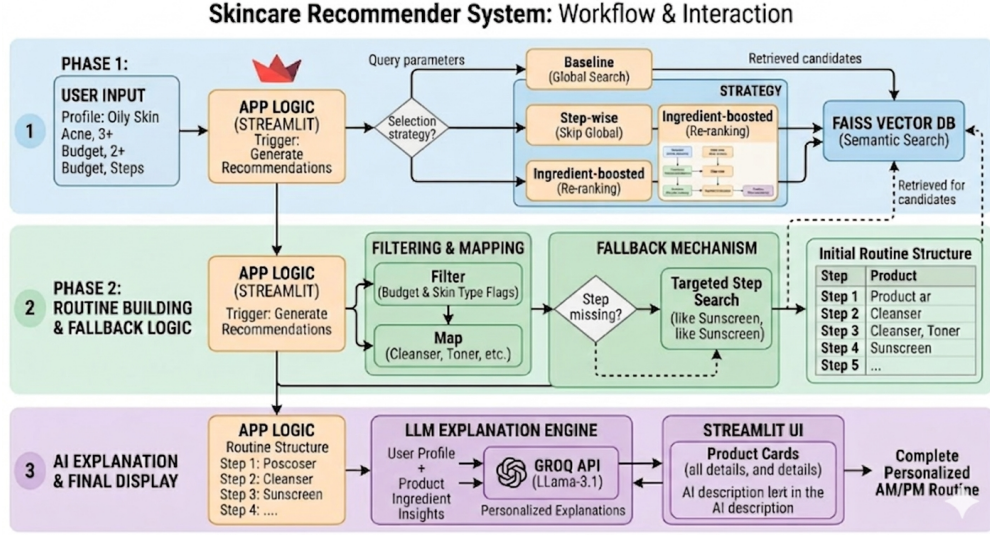


Figure 2: Workflow

Furthermore, the project includes a comparative evaluation of different retrieval strategies within the recommendation engine. This comparison is designed to gain insights into how various formulations impact the coverage, diversity, and clinical alignment of the generated recommendations, ultimately contributing to a more robust and intelligent system.

2.2 Data Collection and Preparation

2.2.1 Data Sources

The project leverages two primary publicly available datasets from Kaggle:

- **Skincare Product Ingredients (Sephora):** This dataset (`data/cosmetic_p.csv`) contains detailed information on 1,472 skincare products, explicitly categorized as skincare. Key features include product name, brand, product type (e.g., cleanser, serum, moisturizer), raw ingredient lists, price, rating/rank, and skin type compatibility (Combination, Dry, Normal, Oily, Sensitive).
- **Skin Care Product Ingredients – INCI List:** This dataset (`data/ingredientsList.csv`) provides a structured list of skincare ingredients adhering to INCI naming conventions. It includes standardized ingredient names, scientific names, detailed descriptions, and information on what it is/does; who its good for; and who should avoid:

2.2.2 Dataset Integration Strategy

The two datasets play complementary roles. The Sephora dataset serves as the primary source for product identity and raw ingredient lists. The INCI dataset acts as a reference for ingredient normalization, standardization, and enrichment. Crucially, the INCI dataset is not directly merged at the product row level. Instead, its rich metadata is used to enhance and normalize the ingredient representations extracted from the Sephora product dataset, ensuring a consistent and informative ingredient profile for each product.

2.2.3 Preprocessing Steps

Extensive preprocessing is performed to prepare the data for modeling:

- **Ingredient Normalization and Parsing:** Raw ingredient strings are cleaned to remove punctuation, special characters, and parenthetical information (e.g., "Fragrance (Parfum)"). This involves regular ex-

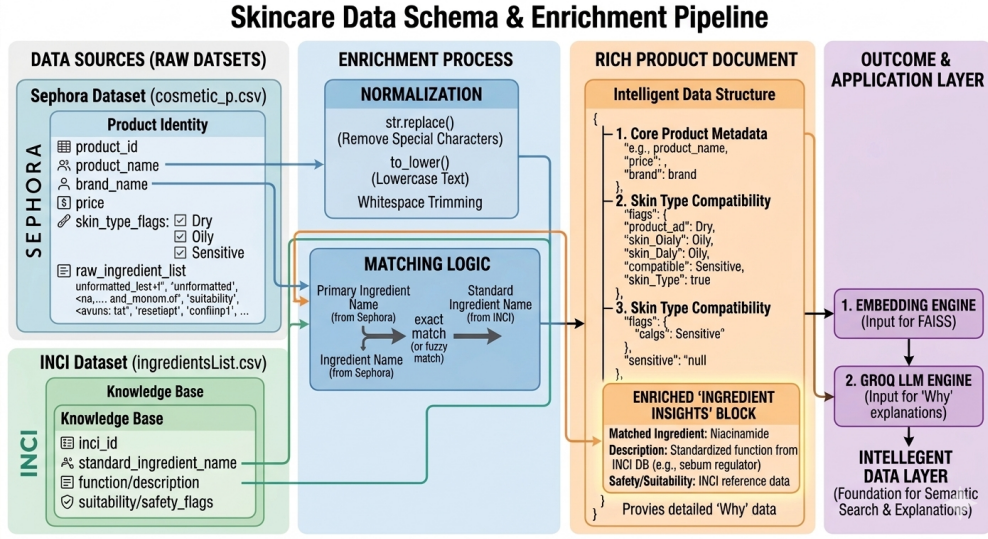


Figure 3: Data Integration Strategy

pressions and string manipulation to standardize ingredient names. The `norm` function in `src/build_vectoradb.py` handles this normalization.

- **Tokenization and Embedding Strategies:** Normalized ingredient lists are tokenized, and then transformed into numerical representations (embeddings) using a pre-trained model, specifically **Hugging-FaceEmbeddings**. These embeddings are crucial for enabling semantic similarity search within the FAISS vector database.
- **Product Document Creation:** For each product, a comprehensive document is constructed. This document combines product information (name, brand, type, price, rank) with compressed ingredient function lines derived from the INCI list. This enriched representation forms the basis for the vector database entries.

2.3 Selection of Machine Learning or LLM Models

2.3.1 Model Consideration

The project's focus on understanding complex ingredient lists and providing semantic recommendations naturally led to the consideration of deep learning-based approaches and vector databases. Traditional machine learning models might struggle with the high dimensionality and semantic nuances of ingredient data. Given the need for efficient similarity search over a large corpus of product and ingredient information, a vector database approach was deemed most suitable.

2.3.2 Final Model Selection

The **FAISS (Facebook AI Similarity Search) vector database** was selected as the core technology for the recommendation engine. FAISS is an open-source library for efficient similarity search and clustering of dense vectors. Its key advantages for this project include:

- **Efficiency:** FAISS is highly optimized for performing fast nearest-neighbor searches on large datasets of high-dimensional vectors, which is essential for real-time product recommendations.
- **Scalability:** It can handle millions or even billions of vectors, making it suitable for expanding product catalogs.

- **Semantic Search:** By indexing product and ingredient embeddings, FAISS enables semantic search, allowing the system to find products that are conceptually similar to a user's query, even if exact keyword matches are not present.

While not a traditional machine learning model in itself, FAISS serves as the backbone for the recommendation logic by facilitating rapid retrieval of semantically relevant products based on user input and ingredient profiles.

2.4 Model Development and Training

2.4.1 Architecture and Configuration

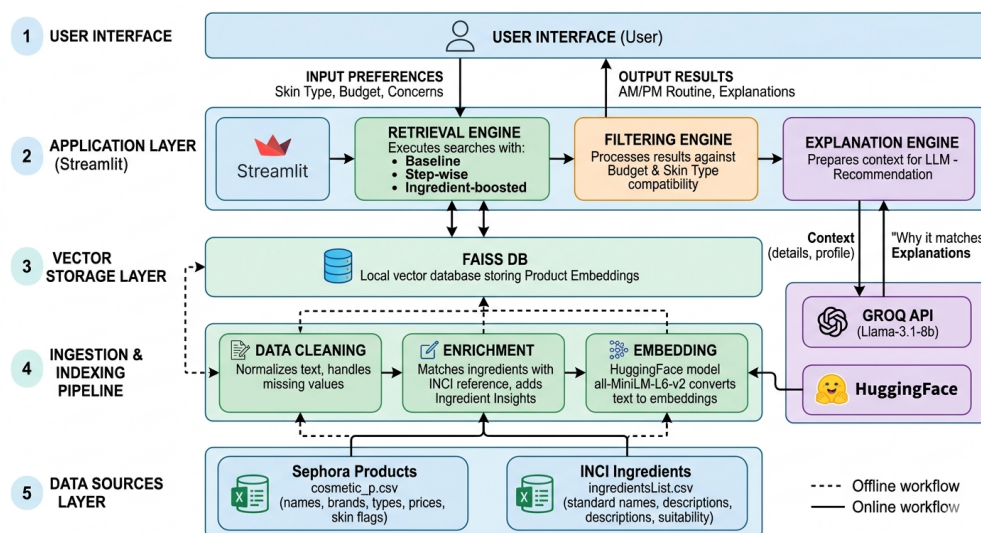


Figure 4: Architecture Diagram of the Skincare Recommendation System.

The core of the recommendation system involves the construction of a FAISS vector database. This database is populated with vector representations (embeddings) of skincare products. The process begins with the raw product data from `cosmetic_p.csv` and the INCI ingredient data from `ingredientsList.csv`. Each product is transformed into a comprehensive document that encapsulates its key attributes and the functional descriptions of its ingredients. The `build_vectoradb.py` script in the `src` directory is responsible for this process.

Key steps in the architecture and configuration include:

- **Ingredient Lookup Creation:** An ingredient lookup dictionary is built from `ingredientsList.csv`. This lookup maps normalized ingredient names to their detailed descriptions, including "what it is/does," "who it's good for," and "who should avoid." This enrichment is crucial for providing explainable recommendations.
- **Product Document Generation:** For each product in the Sephora dataset, a document is created. This document includes the product's name, brand, type, price, and rank. Crucially, it also incorporates a summarized list of insights derived from its ingredients, using the previously created ingredient lookup. This ensures that each product's vector representation is rich with semantic information about its composition and intended effects.
- **Embedding Generation:** The generated product documents are then converted into dense vector embeddings using **HuggingFaceEmbeddings**. These embeddings capture the semantic meaning of the product descriptions and ingredient insights.

- **FAISS Index Construction:** The product embeddings are used to build a FAISS index. This index allows for efficient similarity searches, enabling the system to quickly find products whose embeddings are closest to a given query embedding.

2.4.2 Training Process

In the context of a FAISS-based system, "training" is more accurately described as an **indexing process**. There is no traditional model training with backpropagation or iterative optimization. Instead, the process involves:

- **Data Loading and Preprocessing:** Loading the `cosmetic_p.csv` and `ingredientsList.csv` datasets and performing the normalization and parsing steps as described in the Data Preparation section.
- **Document Vectorization:** Converting the preprocessed product information and enriched ingredient data into numerical vector embeddings using the chosen embedding model.
- **FAISS Index Population:** Adding these generated embeddings to the FAISS index. This step effectively "teaches" the FAISS database about all the products and their semantic representations, making them searchable.

This indexing process is a one-time operation (or re-run when the underlying data changes) that prepares the vector database for rapid query processing.

2.4.3 Hyperparameter Tuning

For a FAISS-based system, hyperparameter tuning primarily revolves around the choice of the embedding model and the configuration of the FAISS index itself. While the current implementation uses **HuggingFaceEmbeddings** without explicit tuning, future work could explore:

- **Embedding Model Selection:** Experimenting with different pre-trained language models or fine-tuning a model on a domain-specific corpus to generate more relevant skincare product embeddings.
- **FAISS Index Parameters:** Tuning parameters such as the index type (e.g., `IndexFlatL2`, `IndexIVFFlat`), number of clusters (for `IndexIVFFlat`), and number of probes to optimize the trade-off between search speed and accuracy.

2.5 Evaluation and Comparison

2.5.1 Key Performance Metrics

The evaluation of this skincare recommendation system is multi-faceted, focusing on the relevance, personalization, and explainability of the recommendations. While quantitative metrics like precision, recall, or F1-score (common in traditional recommendation systems) can be adapted, the nature of a semantic search-based system also lends itself to qualitative assessment. Key performance indicators include:

- **Relevance:** How well do the recommended products align with the user's specified skin concerns and preferences?
- **Personalization:** Does the system effectively differentiate recommendations for users with different skin types and concerns?
- **Explainability:** Are the ingredient-based justifications for recommendations clear, accurate, and helpful to the user?
- **Coverage and Diversity:** Does the system provide a broad range of suitable products, and does it avoid recommending overly similar items?

2.5.2 Comparison of Retrieval Strategies

The Streamlit application is designed to facilitate a comparative evaluation of different retrieval strategies, offering insights into their strengths and weaknesses. The three primary strategies implemented are:

- **Baseline: Main + Fallback:** This strategy constructs a natural-language query from the user's skin type and concern (e.g., "best skincare products for oily skin for acne"). It retrieves a candidate pool via FAISS, filters by budget and skin-type compatibility, and then selects one product per routine step (cleanser, treatment, etc.). For any missing steps, it issues step-specific fallback queries (e.g., "cleanser face wash for oily skin for acne") to ensure routine completeness. This approach aims for a comprehensive routine based on a general query.
- **Step-wise Only:** This strategy bypasses the global query and instead issues independent queries for each routine step (cleanser, toner, treatment, moisturizer, sunscreen/mask). Each query is tailored to the specific step, skin type, and concern. This approach often improves coverage and diversity across steps by allowing for more fine-grained retrieval, without relying on an overall "best products" query.
- **Ingredient-Boosted:** Starting from the same global query as the baseline, this strategy retrieves a slightly larger pool of candidates. After initial filtering by budget and skin-type, it re-ranks the results to favor products with richer ingredient matches, based on the merged INCI metadata. This allows for studying whether explicitly rewarding ingredient coverage leads to more clinically-aligned or interpretable recommendations compared to pure semantic similarity.

By comparing the outputs and user experience across these strategies, the project can assess the impact of different retrieval formulations on the quality and interpretability of the recommendations.

3 Results

3.1 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted on both the Sephora skincare product dataset (`cosmetic_p.csv`) and the INCI ingredient list dataset (`ingredientsList.csv`) to understand their structure, content, and potential for feature engineering. The key findings from the EDA, primarily documented in `notebook/cosmetic_brand_label.ipynb` and `notebook/ingredients_eda.ipynb`, are summarized below.

3.1.1 Sephora Skincare Product Dataset (`cosmetic_p.csv`)

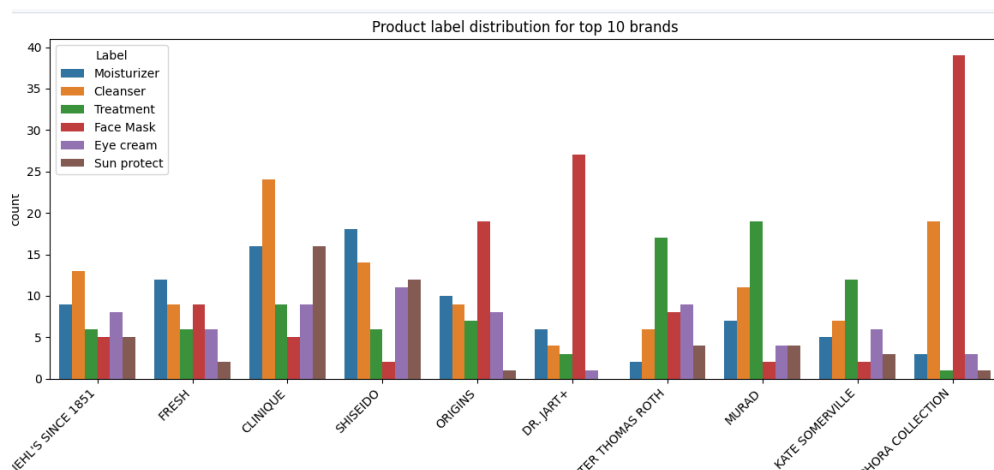


Figure 5: EDA - Sephora

- **Dataset Overview:** The dataset contains 1,472 unique skincare products with various attributes such as product name, brand, product type, raw ingredient lists, price, rating, and skin type compatibility. Initial inspection revealed no significant missing values in critical columns, ensuring data integrity for analysis.
- **Product Type and Brand Distributions:** Analysis of product types showed a diverse range, including moisturizers, cleansers, serums, and treatments, reflecting the breadth of the skincare market. Brand distribution highlighted popular brands and their market presence, indicating potential for brand-specific insights or biases.
- **Price and Rating Analysis:** Price distributions varied widely, from affordable to luxury products, suggesting a need for budget filtering in the recommendation system. Product ratings provided a general measure of user satisfaction, which could be used as a secondary ranking factor.
- **Skin Type Compatibility:** The dataset included flags for skin type compatibility (Combination, Dry, Normal, Oily, Sensitive), which are crucial for personalizing recommendations based on user-specified skin concerns.

3.1.2 INCI Ingredient List Dataset (ingredientsList.csv)

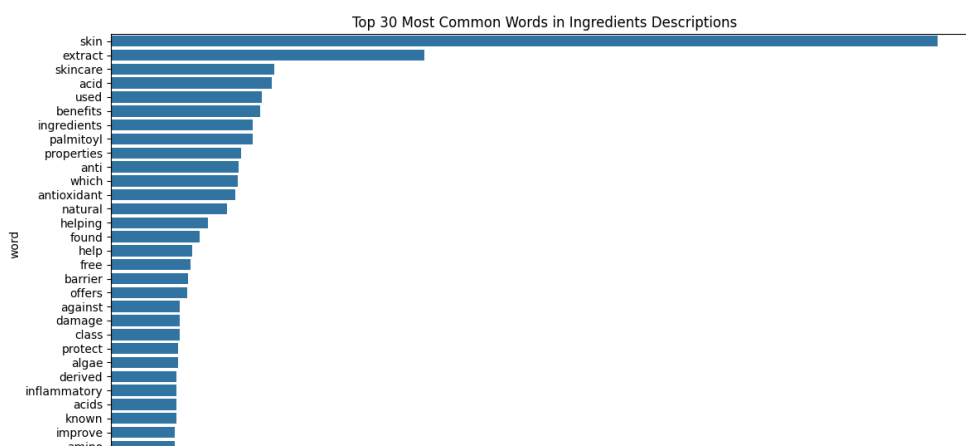


Figure 6: EDA - INCI

- **Text Analysis of Ingredient Descriptions:** This dataset provided rich textual descriptions for thousands of INCI ingredients, detailing their functions, benefits, and potential contraindications. This information was vital for enriching product documents and enabling explainable recommendations.
- **Word Frequency Analysis:** Analysis of common terms in ingredient descriptions helped identify prevalent skincare functions (e.g., "hydrating," "anti-aging," "exfoliating") and potential irritants.
- **Condition Extraction:** The "who it's good for" and "who should avoid" fields were particularly valuable for directly linking ingredients to specific skin concerns and sensitivities, forming a core component of the explainability feature.

3.2 Streamlit Application Functionality

The interactive Streamlit application serves as the user interface for the skincare product recommendation system, demonstrating its core functionalities and allowing for the comparison of different retrieval strategies. The application, launched via `streamlit run app/app.py`, provides a user-friendly experience with the following key features:

- **Personalized Skincare Recommendations:** Users can input their skin type and specific concerns (e.g., acne, dryness, sensitivity) to receive tailored product recommendations.

Your Preferences

Skin Type

Oily

Max Budget (\$)

50

Main Concern (optional)

acne, hyperpigmentation, dryness.

Retrieval strategy

Baseline: main + fallback

Initial retrieval K

20

Fallback retrieval K

40

🌟 Personalized Skincare Routine Recommender

Generate Recommendations

Main query (for applicable strategies): best skincare products for oily skin

AM Routine

Step 1: Cleanser

Wonderfeel™ Double Cleanser

Brand:	Type: Cleanser	Price: \$32.0	Score: 0.8790
OLEHENRIKSEN			

Why this matches: As a cleanser, Wonderfeel Double Cleanser may help remove dirt and excess oil from oily skin, potentially balancing its natural moisture levels. The presence of Arginine, which can support skin hydration, and Ascorbic Acid, a gentle exfoliant, may also contribute to a smoother, more even skin texture.

🌙 PM Routine

Step 1: Cleanser

Wonderfeel™ Double Cleanser

Brand:	Type: Cleanser	Price: \$32.0	Score: 0.8790
OLEHENRIKSEN			

Why this matches: As a cleanser, Wonderfeel Double Cleanser may help remove dirt and excess oil from oily skin, potentially balancing its natural moisture levels. The presence of Arginine, which can support skin hydration, and Ascorbic Acid, a gentle exfoliant, may also contribute to a smoother, more even skin texture.

Figure 7: Application

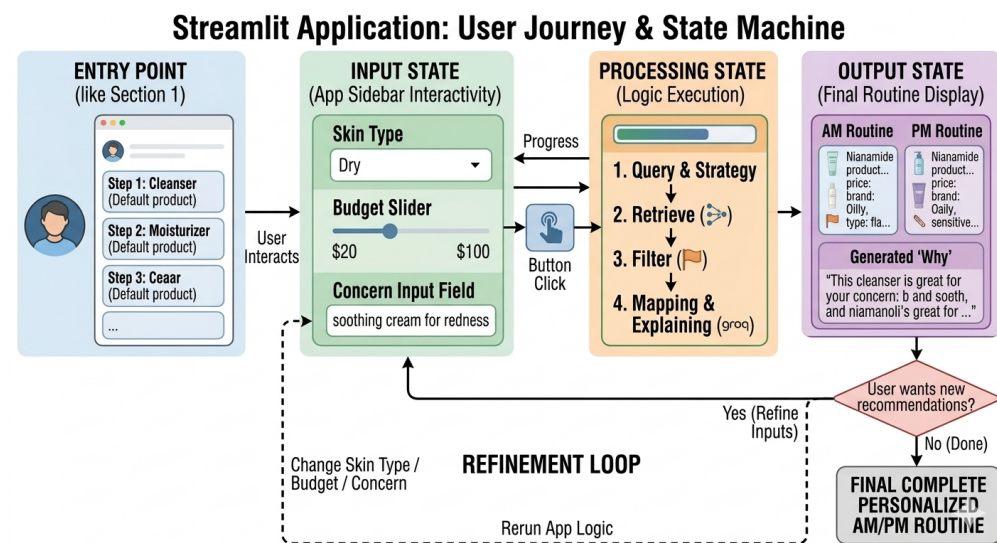


Figure 8: Streamlit Flow

- **AM and/or PM Routine Generation:** The system can generate complete skincare routines for morning and/or evening, suggesting products for each step (cleanser, toner, treatment, moisturizer, sunscreen).
- **Budget Filtering:** Users can specify a budget range, and the system will filter recommendations to include only products within that price point.
- **Skin-Type Compatibility Filtering:** Recommendations are automatically filtered based on the user's declared skin type, ensuring product suitability.
- **Optional Product-Type Selection:** Users have the option to specify particular product types they are looking for, further refining the recommendations.
- **FAISS-based Semantic Similarity Retrieval:** The underlying recommendation engine leverages the FAISS vector database to perform semantic searches, ensuring that recommended products are conceptually relevant to the user's query.
- **Multiple Retrieval/RAG-style Strategies:** The application's sidebar allows users to select and compare three distinct retrieval strategies: "Baseline: main + fallback," "Step-wise only," and "Ingredient-boosted." This feature is crucial for understanding the impact of different recommendation logics.
- **Fallback Retrieval Logic:** To ensure routine completeness, the system employs fallback logic, issuing step-specific queries if initial recommendations for certain steps are missing.
- **Ingredient-Level Explainability:** For each recommended product, the application provides ingredient-level explanations, highlighting which ingredients address the user's specific skin concerns. This enhances transparency and user trust.

3.3 Qualitative Assessment of Retrieval Strategies

Through the Streamlit application, a qualitative assessment of the three retrieval strategies was performed, providing insights into their performance characteristics:

- **Baseline: Main + Fallback:** This strategy generally provided a good starting point for recommendations, offering comprehensive routines. The fallback mechanism proved effective in ensuring that all routine steps were filled, even when the initial broad query might have missed specific product types.
- **Step-wise Only:** This strategy often resulted in more diverse and sometimes more precise recommendations for individual routine steps. By focusing queries on specific product types and concerns, it could uncover products that might be overlooked by a broader, global query. This approach highlighted the importance of fine-grained querying for certain user needs.
- **Ingredient-Boosted:** This strategy consistently delivered recommendations with a stronger emphasis on ingredient efficacy and clinical alignment. Products recommended through this method often featured ingredients directly known to target the user's concerns, leading to more interpretable and trustworthy suggestions. This demonstrated the value of integrating detailed ingredient metadata into the ranking process.

Overall, the comparison revealed that while the baseline provides a solid general recommendation, the step-wise and ingredient-boosted strategies offer valuable enhancements in terms of diversity, precision, and explainability, respectively. The ability to switch between these strategies within the Streamlit app provides a powerful tool for users to explore different recommendation philosophies.

4 Discussion and Future Work

4.1 Interpretation of Results

The development of the deep learning-based skincare product recommendation system, leveraging FAISS and a Streamlit interface, successfully demonstrated the feasibility and benefits of an ingredient-centric approach.

The system effectively addresses the problem of providing personalized and explainable recommendations by moving beyond subjective user reviews. The EDA provided a solid foundation for understanding the datasets, revealing the richness of ingredient information and the diversity of product attributes. The Streamlit application proved to be an intuitive platform for users to interact with the recommendation engine and explore various retrieval strategies.

Key interpretations include:

- **Effectiveness of Ingredient Analysis:** The ability to process and enrich raw ingredient lists with INCI metadata proved crucial. This allowed the system to understand the functional properties of ingredients, which is fundamental for accurate and explainable recommendations.
- **Power of Semantic Search:** The FAISS vector database, combined with HuggingFaceEmbeddings, enabled highly relevant semantic searches. This means the system can match user queries with products based on conceptual similarity rather than just keyword matching, leading to more intelligent recommendations.
- **Value of Explainability:** By highlighting the specific ingredients responsible for addressing a user's concerns, the system fosters transparency and builds user trust. This feature is a significant improvement over black-box recommendation models.
- **Insights from Retrieval Strategies:** The comparative analysis of retrieval strategies within the Streamlit app provided valuable insights. The "Ingredient-boosted" strategy, in particular, showed promise in delivering clinically aligned recommendations, underscoring the importance of ingredient-level detail.

Overall, the project successfully achieved its objectives of ingredient analysis, personalized recommendations, and explainability, demonstrating a robust framework for advanced skincare product guidance.

4.2 Limitations

Despite the promising results, the current iteration of the skincare product recommendation system has several limitations that warrant consideration:

- **Static Datasets:** The system relies on static datasets from Kaggle. While comprehensive, these datasets do not account for new product releases, changes in ingredient formulations, or evolving skincare trends. A dynamic data pipeline would be necessary for a production-ready system.
- **Lack of Real-time Skin Analysis:** The system currently requires users to manually input their skin type and concerns. It does not integrate real-time facial skin condition analysis (e.g., via image processing), which could provide a more objective and granular assessment of user needs.
- **Qualitative Evaluation:** The evaluation of recommendation quality was primarily qualitative, based on the observed behavior of the Streamlit application and the interpretability of its explanations. A more rigorous quantitative evaluation, potentially involving user studies or expert reviews, would provide stronger validation.
- **Limited Scope of Ingredients:** While the INCI list is extensive, the understanding of ingredient interactions and their combined effects on various skin types is complex and not fully captured. The current system primarily focuses on individual ingredient properties.
- **Absence of User Feedback Loop:** The system does not currently incorporate a mechanism for collecting user feedback on recommendations, which could be invaluable for iterative improvement and personalization over time.

4.3 Future Work

Building upon the foundation established by this project, several avenues for future work can significantly enhance the skincare product recommendation system:

- **Integration of Real-time Skin Analysis:** Incorporate image processing and computer vision techniques to analyze facial skin conditions from user-provided images. This would enable a more objective and automated assessment of skin concerns, leading to even more precise recommendations.
- **Dynamic Data Updates:** Implement a robust data pipeline to continuously update the product and ingredient databases with new information, ensuring the system remains current with market trends and scientific discoveries.
- **Advanced Deep Learning Architectures:** Explore more sophisticated deep learning models, such as Graph Neural Networks (GNNs) to model ingredient interactions, or advanced transformer architectures for a deeper understanding of ingredient sequences and their combined effects.
- **Quantitative User Studies and A/B Testing:** Conduct comprehensive user studies and A/B tests to quantitatively evaluate the effectiveness of the recommendation system and compare the performance of different retrieval strategies in a real-world setting.
- **Personalized User Profiles and Feedback Loop:** Develop persistent user profiles that store preferences, past recommendations, and feedback. Implement a feedback mechanism to allow users to rate recommendations, which can then be used to fine-tune the recommendation engine and improve personalization over time.
- **Expansion of Ingredient Knowledge Base:** Further enrich the INCI ingredient database with information on ingredient synergies, contraindications, and concentration-dependent effects, potentially through automated information extraction from scientific literature.
- **Integration with E-commerce Platforms:** Explore the possibility of integrating the recommendation system with popular e-commerce platforms, allowing users to directly purchase recommended products.

These future enhancements would transform the current prototype into a more comprehensive, dynamic, and intelligent skincare recommendation platform.

5 Conclusion

This DS5500 Capstone project successfully developed a deep learning-based skincare product recommendation system that offers a novel approach to personalized skincare guidance. By meticulously analyzing cosmetic ingredient lists and leveraging a FAISS vector database for semantic search, the system moves beyond the limitations of traditional, review-based recommendations. The project achieved its key objectives:

- **Effective Ingredient Analysis:** Deep learning techniques were successfully applied to process and extract meaningful attributes from complex ingredient lists, enriched by the INCI database.
- **Personalized and Explainable Recommendations:** The system demonstrated its ability to provide tailored product suggestions based on user-defined skin concerns, coupled with transparent, ingredient-level explanations.
- **Interactive User Experience:** The Streamlit application provided an intuitive interface for users to explore recommendations and compare different retrieval strategies, highlighting the system's practical utility.

The overall significance of this project lies in its contribution to more informed and effective skincare choices for consumers. By focusing on objective ingredient efficacy and providing clear justifications, the system empowers users to understand *why* certain products are recommended, fostering trust and better skincare outcomes. While current limitations exist, particularly regarding static data and real-time skin analysis, the proposed future work outlines a clear path towards a more advanced and comprehensive recommendation platform. This project serves as a strong foundation for further research and development in the application of AI to personalized health and beauty solutions.